

Discrete
Mathematics
Segment Three



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1. Introduction to function

Let A and B be nonempty sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f : A \rightarrow B$.

Functions are sometimes also called **mappings** or **transformations**

1. Introduction to function

If f is a function from A to B , we say that A is the *domain* of f and B is the *codomain* of f . If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b . The *range*, or *image*, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f maps A to B .

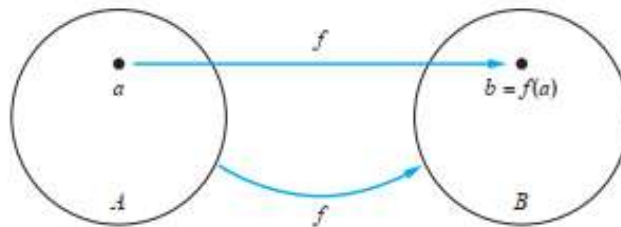


FIGURE 2 The Function f Maps A to B .

1. Introduction to function

Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$

$$(f_1 f_2)(x) = f_1(x) f_2(x).$$

Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

One-to-One Function | Injective

A function f is said to be *one-to-one*, or an *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be *injective* if it is one-to-one.

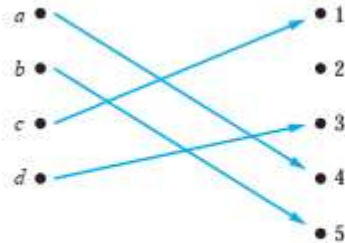


FIGURE 3 A One-to-One Function.

One-to-One Function | Injective

EXAMPLE 8 Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.



Solution: The function f is one-to-one because f takes on different values at the four elements of its domain. This is illustrated in Figure 3. 

EXAMPLE 9 Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution: The function $f(x) = x^2$ is not one-to-one because, for instance, $f(1) = f(-1) = 1$, but $1 \neq -1$.

Onto Function | Surjective

A function f from A to B is called *onto*, or a *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called *surjective* if it is onto.

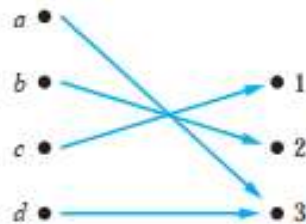


FIGURE 4 An Onto Function.

Onto Function | Surjective

EXAMPLE 12 Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?



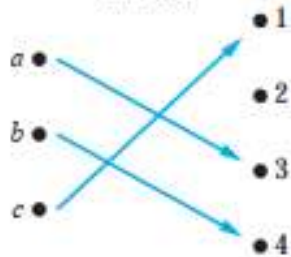
Solution: Because all three elements of the codomain are images of elements in the domain, we see that f is onto. This is illustrated in Figure 4. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto. ◀

EXAMPLE 13 Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

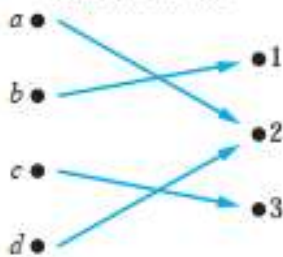
Solution: The function f is not onto because there is no integer x with $x^2 = -1$, for instance. ◀

Different Types of Correspondences

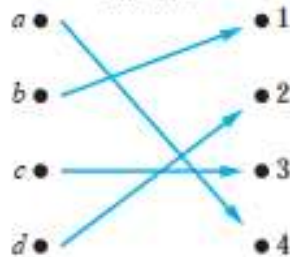
(a) One-to-one,
not onto



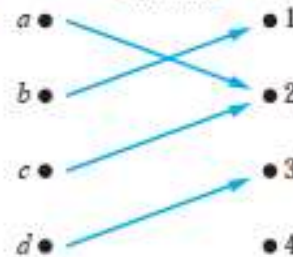
(b) Onto,
not one-to-one



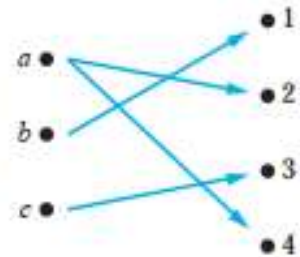
(c) One-to-one,
and onto



(d) Neither one-to-one
nor onto



(e) Not a function



Bijjective

The function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto. We also say that such a function is *bijjective*.

EXAMPLE 16 Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijection?

Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection. ◀

Exercise

Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$ that is:

- (i) One-one but not onto**
- (ii) One-one and onto**
- (iii) Neither one-one nor onto**
- (iv) Onto but not one-one**

Exercise

(i) One-one but not onto

Ans:

Let the function $f : N \rightarrow N$, given by $f(x) = 2x$.

Let us consider two elements x_1 and x_2 , in the domain of 'f'. So we get –

$$f(x_1) = 2x_1 \text{ and } f(x_2) = 2x_2.$$

Now, we know condition for one-one is given as –

$$f(x_1) = 2x_1 \text{ and } f(x_2) = 2x_2.$$

Now, putting $f(x_1) = f(x_2)$, we get

$$f(x_1) = f(x_2)$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Hence, the given function is One-one.

Now Let $f(x) = y$, such that $y \in N$

We get –

$$2x = y$$

$$x = \frac{y}{2}$$

Put $y = 1$

$$x = \frac{1}{2} = 0.5, \text{ which cannot be true as } x \in N \text{ as supposed in solution.}$$

Hence, the given function is not onto.

So, $f(x) = 2x$ is an example of One-one but not onto function.

Exercise

(ii) One-one and onto

Ans:

Suppose $f(x) = x$, for $x : R \rightarrow R$

Calculate $f(x_1)$ and $f(x_2)$ as $f(x_1) = x_1$, and $f(x_2) = x_2$.

Now, equating $f(x_1)$ and $f(x_2)$, we get

$$f(x_1) = f(x_2)$$

$$x_1 = x_2$$

Hence, $f(x) = x$ for $x : R \rightarrow R$ is a one-one function.

Now, suppose $f(x) = y$, where $y \in R$, $x = y$.

Now, for all $y \in R$, $x \in R$, as well. Hence, $f(x) = x$ is onto function as well.

So, $f(x) = x$, given as $x : R \rightarrow R$ is one-one and onto both.

Exercise

(iii) Neither one-one nor onto

(iii) Let function $f : R \rightarrow R$, given by $f(x) = x^2$

Calculate $f(x_1)$:

$$f(x_1) = x_1^2$$

$$f(x_2) :$$

$$f(x_2) = x_2^2$$

Now, $f(x_1) = f(x_2)$

$$x_1^2 = x_2^2$$

$$x_1^2 - x_2^2 = 0$$

We know the algebraic identity of $a^2 - b^2$ can be given as –

$$a^2 - b^2 = (a - b)(a + b),$$

So, we get –

$$(x_1 - x_2)(x_1 + x_2) = 0.$$

$$x_1 - x_2 = 0 \text{ or } x_1 + x_2 = 0 \text{ or } x_1 = -x_2 \text{ or } x_1 = x_2.$$

Since, x , does not have a unique image as per the above statements, hence, it is not one-one.

$$\text{Now, } f(x) = x^2$$

Let $f(x) = y$, such that $y \in R$.

$$y = x^2$$

$$x = \pm\sqrt{y}$$

Put $y = -5$, we get –

$$x = \pm\sqrt{-5}$$

Where, x is not a real value. So, $f(x) = x^2$ is not one-one and not onto as well for $x : R \rightarrow R$.

Exercise

(iv) Onto but not one-one

(iv) Let the function $f : N \rightarrow N$, given by $f(1) = f(2) = 1$

Here, $f(x) = f(2) = 1$ and $f(x) = f(2) = 1$

Since, different elements 1 and 2 have the same image '1'. Hence, it cannot be a one-one function.

Let $f(x) = y$, such that $y \in N$.

Here, y is a natural number for every 'y', there is a value of x which is a natural number.

Hence, f is onto.

So, the function $f : N \rightarrow N$, given by $f(1) = f(2) = 1$ is not one-one but onto.

Exercise

Question: Write down the difference between one-to-one and onto function with proper examples. [Aut'23]

Answer:

A **one-to-one** function (injective) is a function in which no two different elements in the domain map to the same element in the codomain.

Example: $f(x) = 2x$ is one-to-one.

An **onto function** (surjective) is a function in which every element in the codomain is mapped to by at least one element in the domain.

Example: $f(x) = x^2$ is not onto because it misses negative numbers in the codomain.

Exercise

Consider these functions from the set of students in a discrete mathematics class. Under what conditions is the function one-to-one if it assigns to a student his or her

- | | |
|-----------------------------|-----------------------------------|
| a) mobile phone number. | b) student identification number. |
| c) final grade in the class | d) Using class room |
| e) home town | f) Use computer of lab |

Solution:

- a) If there are no two students with a common phone, the function that assigns to a student his or her mobile phone number is one-to-one.
- b) Since the student identification number is unique, and hence there are no two students with the same identification number, the function that assigns to a student his or her identification number is one-to-one.
- c) Two student can have the same final grade in the class. Therefore, in general case this function is not one-to-one. It can be one-to-one if the number of students is less or equal to the number of final grades and different students have different final grades.
- d) This function in general case is not one-to-one because of all students of a section use a single classroom for a lecture.
- e) This function in general case is not one-to-one because of two students can live in the same town. It can be injective if different students have different home town.
- f) If there are no two students with a common configured computer, the function that assigns to use a computer of lab is one-to-one.

Inverse Function

Let f be a one-to-one correspondence from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

A one-to-one correspondence is called **invertible** because we can define an inverse of this function. A function is not invertible if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

Composition Function

Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The *composition* of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is defined by

$$(f \circ g)(a) = f(g(a)).$$

In other words, $f \circ g$ is the function that assigns to the element a of A the element assigned by f to $g(a)$. That is, to find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$.

Composition Function

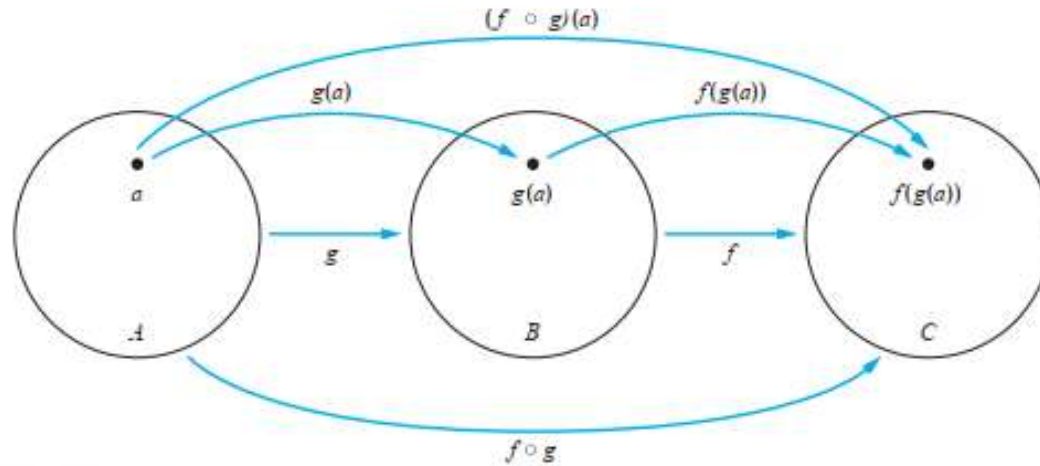


FIGURE 7 The Composition of the Functions f and g .

Composition Function

Example: Let $A = \{1, 2, 3, 4\}$, $f = \{(1, 4), (2, 1), (3, 3), (4, 2)\}$ and $g = \{(1, 3), (2, 1), (3, 2), (4, 4)\}$; find (i) gof (ii) fog (iii) fof

Answer:

$$(i) \quad \text{gof}(1) = g(f(1)) = g(4) = 4$$

$$\text{gof}(2) = g(f(2)) = g(1) = 3$$

$$\text{gof}(3) = g(f(3)) = g(3) = 2$$

$$\text{gof}(4) = g(f(4)) = g(2) = 1$$

$$\text{So,} \quad \text{gof} = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$

$$\text{Similarly} \quad \text{fog} = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$$

$$\text{fof} = \{(1, 2), (2, 4), (3, 3), (4, 1)\}$$

Composition Function

Example If $f(x) = x + 7$ and $g(x) = x - 7$, $x \in R$, find $(f \circ g)(7)$

Solution. We have, $f(x) = x + 7$ and $g(x) = x - 7$

$$\begin{aligned}\text{Now } (f \circ g)(7) &= f[g(7)] \\ &= f[7 - 7] \\ &= f(0) \\ &= 0 + 7 = 7\end{aligned}$$

Composition Function

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? g and f ?

Ans: Given: $f(x) = 2x + 3$

$$g(x) = 3x + 2$$

to find: $f(g(x))$ and $g(f(x))$

solution: 1. $f(g(x))$ simply means the value of $g(x)$ in the variable of $f(x)$

therefore, putting $3x + 2$ in place of x (variable) of $f(x)$

$$\text{i.e. } 2(3x + 2) + 3 = 6x + 7$$

2. $g(f(x))$ simply means the value of $f(x)$ in the variable of $g(x)$

therefore, putting $2x + 3$ in place of x (variable) of $g(x)$

$$\text{i.e. } 3(2x + 3) + 2 = 6x + 11$$

Composition Function

Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements. [Aut'23]

- i) If $f \circ g$ is onto, then f must also be onto.
- ii) If $f \circ g$ is one-to-one, then g must also be one-to-one.

i) Answer:

Let's consider two functions:

Function $g: \{1, 2, 3\} \rightarrow \{a, b, c\}$ defined as $g(1) = a, g(2) = b, g(3) = c$.

Function $f: \{a, b, c\} \rightarrow \{X, Y, Z\}$ defined as $f(a) = X, f(b) = Y, f(c) = Z$.

Now, let's compute $f \circ g$:

$$(f \circ g)(1) = f(g(1)) = f(a) = X$$

$$(f \circ g)(2) = f(g(2)) = f(b) = Y$$

$$(f \circ g)(3) = f(g(3)) = f(c) = Z$$

$f \circ g$ is onto because for every element in the codomain of $f \circ g$ ($\{X, Y, Z\}$), there exists an element in the domain of $f \circ g$ ($\{1, 2, 3\}$) that maps to it.

Now, let's prove that f is also onto:

For every element in the codomain of f ($\{X, Y, Z\}$), there exists an element in the domain of f ($\{a, b, c\}$) that maps to it.

$$f(a) = X$$

$$f(b) = Y$$

$$f(c) = Z$$

Therefore, f is onto as well.

Composition Function

Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements.

[Aut'23]

- i) If $f \circ g$ is onto, then f must also be onto.
- ii) If $f \circ g$ is one-to-one, then g must also be one-to-one.

ii) Answer:

Let's consider two functions:

Function $g: \{1, 2, 3\} \rightarrow \{a, b, c\}$ defined as $g(1) = a, g(2) = b, g(3) = c$.

Function $f: \{a, b, c\} \rightarrow \{X, Y, Z\}$ defined as $f(a) = X, f(b) = Y, f(c) = Z$.

Now, let's compute $f \circ g$:

$$(f \circ g)(1) = f(g(1)) = f(a) = X$$

$$(f \circ g)(2) = f(g(2)) = f(b) = Y$$

$$(f \circ g)(3) = f(g(3)) = f(c) = Z$$

$f \circ g$ is one-to-one because, for every pair of distinct elements in the domain of $f \circ g$ ($\{1, 2, 3\}$), their images under $f \circ g$ are also distinct.

Now, let's prove that g is also one-to-one:

For every pair of distinct elements in the domain of g ($\{1, 2, 3\}$), their images under g are also distinct.

$$g(1) = a$$

$$g(2) = b$$

$$g(3) = c$$

Therefore, g is one-to-one as well.

Floor and Ceiling Function

The *floor function* assigns to the real number x the largest integer that is less than or equal to x . The value of the floor function at x is denoted by $\lfloor x \rfloor$. The *ceiling function* assigns to the real number x the smallest integer that is greater than or equal to x . The value of the ceiling function at x is denoted by $\lceil x \rceil$.

EXAMPLE 26 These are some values of the floor and ceiling functions:

$$\lfloor \tfrac{1}{2} \rfloor = 0, \lceil \tfrac{1}{2} \rceil = 1, \lfloor -\tfrac{1}{2} \rfloor = -1, \lceil -\tfrac{1}{2} \rceil = 0, \lfloor 3.1 \rfloor = 3, \lceil 3.1 \rceil = 4, \lfloor 7 \rfloor = 7, \lceil 7 \rceil = 7.$$

Relation

Definition: A **Relation** R from set A to set B is a subset of $A \times B$.

- ❖ If $(a, b) \in R$, we say that " a is related to b ", and write aRb .
- ❖ If $(a, b) \notin R$, we say that " a is not related to b ", and write $a \not R b$.
- ❖ If $A = B$, we often say that $R \in A \times A$ is a relation on A .

Example: $A = (1, 2, 3)$ and $B = \{x, y, z\}$, and let

$$R = \{(1, y), (1, z), (3, y)\}.$$

Then R is a relation from A to B ? . **Yes--since R is a subset of $A \times B$**

With respect to this relation,

$$1Ry, 1Rz, 3Ry, \quad \text{but} \quad 1\cancel{R}x, 2\cancel{R}x, 2\cancel{R}y, 2\cancel{R}z, 3\cancel{R}x, 3\cancel{R}z$$

Relation

Example: Consider these relations on the set of integers:

$$R_1 = \{(a,b) \mid a \leq b\},$$

$$R_4 = \{(a,b) \mid a = b\},$$

$$R_2 = \{(a,b) \mid a > b\},$$

$$R_5 = \{(a,b) \mid a = b + 1\},$$

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\},$$

$$R_6 = \{(a,b) \mid a + b \leq 3\}.$$

Which of these relations contain each of the pairs

$(1,1)$, $(1, 2)$, $(2, 1)$, $(1, -1)$, and $(2, 2)$?

Solution: Note that these relations are on an infinite set and each of these relations is an infinite set. Checking the conditions that define each relation, we see that

$(1,1)$ is in R_1 , R_3 , R_4 , and R_6 :

$(1,2)$ is in R_1 and R_6 :

$(2,1)$ is in R_2 , R_5 , and R_6 :

$(1, -1)$ is in R_2 , R_3 , and R_6 :

$(2,2)$ is in R_1 , R_3 , and R_4 .

Relation

A person deposits \$1000 in an account that yields 9% interest compounded annually. Set up a recurrence relation for the amount in the account at the end of n years. [Aut'23]

Answer:

To set up a recurrence relation for the amount in the account at the end of n years with 9% interest compounded annually, we can use the formula for compound interest:

$$A = P(1 + r/n)^{nt}$$

In this case, we have:

$P = \$1000$ (the principal amount)

$r = 0.09$ (9% annual interest rate in decimal form)

$n = 1$ (compounded annually)

$t = n$ years, so $t = n$

We want to find $A(n)$, which is the amount at the end of n years.

$$A(n) = P(1 + r/n)^{nt}$$

Substituting the given values:

$$A(n) = 1000(1 + 0.09/1)^{(1*n)}$$

Simplify this expression:

$$A(n) = 1000(1 + 0.09)^n$$

Now, the recurrence relation can be expressed as:

$$A(n) = 1000(1.09)^n$$

This recurrence relation represents the amount in the account at the end of n years with 9% interest compounded annually. We can use this relation to calculate the amount for any given n .

Domain & Range

Definition: The **domain** of relation R is the set of all first elements of the ordered pairs which belong to R , denoted by $\text{Dom}(R)$.

Definition: The **range** is the set of second elements of the ordered pairs which belong to R , denoted by $\text{Ran}(R)$.

Example: $A = (1, 2, 3)$ and $B = \{x, y, z\}$, and consider the relation
$$R = \{(1, y), (1, z), (3, y)\}.$$

Find the domain and range of R .

The domain of R is $\text{Dom}(R) = \{1, 3\}$

The range of R is $\text{Ran}(R) = \{y, z\}$

Domain & Range

Example: Find the domain and range of these functions.

- a) the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string
- b) the function that assigns to each bit string twice the number of zeros in that string.
- c) the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)
- d) the function that assigns to each positive integer the largest perfect square not exceeding this integer.
- e) the function that assigns to each pair of positive integers the first integer of the pair.
- f) the function that assigns to each positive integer its largest decimal digit.

Answer:

- a) Domain is the set of all bit strings, range is $\mathbb{Z}$.
- b) Domain is the set of all bit strings, range is the set of even natural number.
- c) Domain is the set of all bit strings, range is $\{0,1,2,3,4,5,6,7\}$
- d) Domain is the set of positive integers, range is $\{1,4,9,16,25,36,\dots\}$
- e) Domain: $\{1,2,3,\dots\}$ Range: $\{1,2,3,\dots\}$
- f) Domain: $\{1,2,3,\dots\}$, Range: $\{0,1,2,3,4,5,6,7,8,9\}$

Domain & Range

Example: Find out these functions' domain and range. To find the domain in each scenario, identify the collection of elements that the function assigned values to.

- a) the function that assigns to each nonnegative integer its last digit. [Aut'23]
- b) the function that assigns the next largest integer to a positive integer. [Aut'23]
- c) the function that assigns to a bit string the number of one bits in the string.
- d) the function that assigns to a bit string the number of bits in the string.

Answer:

a) Domain: The domain is the set of nonnegative integers, which is $\{0, 1, 2, 3, \dots\}$.

Range: The range is the set of possible last digits, which is $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

b) Domain: The domain is the set of positive integers, which is $\{1, 2, 3, 4, \dots\}$.

Range: The range is the set of positive integers that are one greater than the input, which is $\{2, 3, 4, 5, \dots\}$.

c) The domain is the set of all bit strings, and the range is the set of nonnegative integers.

d) The domain is the set of all bit strings, and the range is the set of nonnegative integers (a bit string can have length 0).

Inverse Relations

Definition: Let R be any relation from set A to B . The inverse of R , denoted by R^{-1} , is the relation from B to A denoted by

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

Example: let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Find the inverse of

$$R = \{(1, y), (1, z), (3, y)\}$$

Solution: $R^{-1} = \{(y, 1), (z, 1), (y, 3)\}$

- ❖ If R is any relation, then $(R^{-1})^{-1} = R$.
- ❖ The domain and range of R^{-1} are equal to the range and domain of R , respectively.
- ❖ If R is a relation on A , then R^{-1} is also a relation on A .

Types of Relations

Types of relations which are defined on a set A .

- ❖ Reflexive and Irreflexive Relations
- ❖ Symmetric and Antisymmetric Relations
- ❖ Transitive Relations

Reflexive Relations

A relation R on a set A is called *reflexive* if $(a, a) \in R$ for every element $a \in A$.

EXAMPLE 7 Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

Solution: The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive because $(3, 3)$ is not in any of these relations. ◀

Reflexive Relations

Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of the relations are reflexive?

Solution: R_1 , R_3 , and R_4 are reflexive relations

Symmetric and Antisymmetric Relations

A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$. A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.

EXAMPLE 7 Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of the relations are symmetric and which are antisymmetric?

Ans: R_2 and R_3 are symmetric. R_4, R_5 and R_6 are antisymmetric.

Symmetric and Antisymmetric Relations

Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of the relations are symmetric and which are antisymmetric?

Solution: R_3 , R_4 , and R_6 are symmetric. R_1 , R_2 , R_4 and R_5 are antisymmetric.

Transitive Relations

A relation R on a set A is called *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

EXAMPLE 7 Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of the relations are Transitive?

Solution: R_4 , R_5 , and R_6 are Transitive.

Transitive Relations

Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of the relations are transitive?

Solution: R_1 , R_2 , R_3 and R_4 are transitive.

Types of Relations | Exercise

For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.

- a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$ [Aut'23]
- b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ [Aut'23]
- c) $\{(2, 4), (4, 2)\}$ [Aut'23]
- d) $\{(1, 2), (2, 3), (3, 4)\}$
- e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ [Aut'23]
- f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

Types of Relations | Exercise

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

Answer:

Relation R is not reflexive: $(1,1), (4,4) \notin R$

Relation R is not symmetric: $(2,4) \in R, (4,2) \notin R$

Relation R is not antisymmetric: $(2,3), (3,2) \in R$

Relation R is transitive:

$$(2,2), (2,3) \in R \rightarrow (2,3) \in R;$$

$$(2,2), (2,4) \in R \rightarrow (2,4) \in R;$$

$$(2,3), (3,2) \in R \rightarrow (2,2) \in R;$$

$$(2,3), (3,3) \in R \rightarrow (2,3) \in R;$$

$$(2,3), (3,4) \in R \rightarrow (2,4) \in R;$$

$$(3,2), (2,2) \in R \rightarrow (3,2) \in R;$$

$$(3,2), (2,3) \in R \rightarrow (3,3) \in R;$$

$$(3,2), (2,4) \in R \rightarrow (3,4) \in R;$$

$$(3,3), (3,2) \in R \rightarrow (3,2) \in R;$$

$$(3,3), (3,4) \in R \rightarrow (3,4) \in R$$

Types of Relations | Exercise

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

Answer:

Relation R is reflexive: $(1,1), (2,2), (3,3), (4,4) \in R$

Relation R is symmetric: $(1,2), (2,1) \in R$

Relation R is not antisymmetric: $(1,2), (2,1) \in R$

Relation R is transitive:

$(1,1), (1,2) \in R \rightarrow (1,2) \in R;$

$(2,1), (1,2) \in R \rightarrow (2,2) \in R;$

$(1,2), (2,1) \in R \rightarrow (1,1) \in R;$

$(1,2), (2,2) \in R \rightarrow (1,2) \in R;$

$(2,2), (2,1) \in R \rightarrow (2,1) \in R$

Types of Relations | Exercise

c) $\{(2, 4), (4, 2)\}$

Answer:

Relation R is not reflexive: $(1,1), (2,2), (3,3), (4,4) \notin R$

Relation R is symmetric: $(2,4), (4,2) \in R$

Relation R is not antisymmetric: $(2,4), (4,2) \in R$

Relation R is not transitive:

$(2,4), (4,2) \in R, (2,2) \notin R$	
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Types of Relations | Exercise

d) $\{(1, 2), (2, 3), (3, 4)\}$

Answer:

Relation R is not reflexive: $(1,1), (2,2), (3,3), (4,4) \notin R$

Relation R is not symmetric: $(1,2) \in R, (2,1) \notin R$

Relation R is antisymmetric: $(2,1), (3,2), (4,3) \notin R$

Relation R is not transitive:

$(1,2), (2,3) \in R, (1,3) \notin R$	
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Types of Relations | Exercise

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

Answer:

Relation R is reflexive: $(1,1), (2,2), (3,3), (4,4) \in R$

Relation R is symmetric: $(1,1), (2,2), (3,3), (4,4) \in R$

Relation R is antisymmetric: $(1,1), (2,2), (3,3), (4,4) \in R$

Relation R is transitive.

Types of Relations | Exercise

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

Answer:

Relation R is not reflexive: $(1,1), (2,2), (3,3), (4,4) \notin R$

Relation R is not symmetric: $(1,4) \in R, (4,1) \notin R$

Relation R is not antisymmetric: $(1,3), (3,1) \in R$

Relation R is not transitive:

$(1,3), (3,1) \in R, (1,1) \notin R$	
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Representing Relations

A relation between finite sets can be represented using a **zero-one matrix**.

The **zero-one** matrix representing R has a 1 as its (i, j) entry when a_i is related to b_j , and a 0 in this position if a_i is not related to b_j .

Example: Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A, b \in B$, and $a > b$. What is the matrix representing R if $a_1 = 1, a_2 = 2$, and $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?

Solution: Because $R = \{(2, 1), (3, 1), (3, 2)\}$, the matrix for R is

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

The 1s in M_R show that the pairs $(2, 1)$, $(3, 1)$, and $(3, 2)$ belong to R .
The 0s show that no other pairs belong to R .

Representing Relations

Example: Given sets $A=\{1,2,3,4\}$ and $B=\{2,4,6,8\}$, determine the zero-one matrices that represent the following relations R_1 to R_4 : **[Aut'23]**

i) $R_1=\{(1,2),(2,4),(3,6)\}$

ii) $R_2=\{(1,1),(2,4),(3,9)\}$

iii) $R_3=\{(2,3),(4,6),(3,9)\}$

iv) $R_4=\{(1,2),(2,4),(2,6)\}$

Solution:

i)	ii)	iii)	iv)
1000	0000	0000	1000
0100	0100	0000	0110
0010	0000	0000	0000
0000	0000	0000	0000